Set

Def.: A set is a collection of objects (elements)

$x \in A$: x is an ele. of A

Subset

Def.: A set A is a subset of a set B $(A \subseteq B)$

$$\Leftarrow \forall a \in A,\ a \in B$$

$$\Leftrightarrow \forall x,\ x \in A \Rightarrow x \in B,$$

$$A \not\subseteq B \Leftarrow \exists a \in A,\ \text{s.t.}\ a \notin B$$

$$A = B \Leftarrow \begin{cases} A \subseteq B \\ B \subseteq A \end{cases}$$

Roster notation

$$A = \{a, b, c\} = \{c, b, a\}$$

$$\mathbb{N} = \{0, 1, 2, \cdots, n, \cdots\}$$

① ②

$P(x)$: a statement for (∀) any x

$P(x)$, $x \in D$

$A = \{x \in D : P(x)\} = \{x \in D \mid P(x)\}$

E.g.: $\{n \in \mathbb{Z} : 7 \mid n\} = \{0, \pm 7, \pm 14, \cdots\}$

$\mathbb{R}$: real

$\mathbb{Z}$: int.

$\mathbb{Q}$: rational

$\mathbb{R} \setminus \mathbb{Q}$: irr.

$\mathbb{Z}_{\geq m} = \{n \in \mathbb{Z} : n \geq m\}$

$\mathbb{Z}_{>m}$, $\mathbb{Z}_{\leq m}$, $\mathbb{Z}_{<m}$

Intervals on $\mathbb{R}$

$(a, b) = \{x \in \mathbb{R} : a < x < b\}$

$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$

⋮

Finite set

Def.: A set A is <u>finite</u>
$\Leftarrow$ A has finite many elements

$|A| = n \Leftarrow$ $\exists$ distinct $a_1, \cdots, a_n$, $A = \{a_1, \cdots, a_n\}$
(set)

$A = \emptyset \Leftarrow$ set A has <u>no</u> element

Cardinality (基數)

Def.: The <u>cardinality</u> of a finite set A : $|A|$
is the number of elements of A

$\forall$ finite sets A and B, $A \subseteq B \Rightarrow |A| \leq |B|$

Thm.: ① $|A| = 0 \Rightarrow A = \emptyset$

② $\forall$ set A, $\begin{cases} \emptyset \subseteq A \\ A \subseteq A \end{cases}$

③ $\forall$ set A, B, C, $\begin{cases} A \subseteq B \\ B \subseteq C \end{cases} \Rightarrow A \subseteq C$

Proof: (for ②)

Assume that $\emptyset \notin A$

Then $\exists x \in \emptyset$, s.t. $x \notin A$

~~contradicts: $\forall x, x \notin \emptyset$~~ ✗

$\Rightarrow |\emptyset| \geq 1$, contradicts $|\emptyset| = 0$

② uniqueness: (of $\phi_{\text{②}}$)

let ϕ_1, ϕ_2 be sets s.t. $|\phi_1| = 0$ and $|\phi_2| = 0$

Then $\phi_1 \subseteq \phi_2$ and $\phi_2 \subseteq \phi_1$

$\Rightarrow \phi_1 = \phi_2$

Prove: $\{n \in \mathbb{Z} : n = 2k+1$ for some integer $k\}$

$= \{n \in \mathbb{Z} : \exists k \in \mathbb{Z}, \text{ s.t. } n = 2k-1\}$

Proof:

(②中)

Let $n \in A$,

then $\exists k \in \mathbb{Z}$, s.t. $n = 2k+1$

$= 2(k+1)-1$

let $m = k+1$, $k \in \mathbb{Z} \Rightarrow m \in \mathbb{Z}$

so $n = 2(k+1)-1 = 2m-1 \in B$

Hence $A \subseteq B$

Let $n_2 \in B$,

、 — — (R)

$A = \{x \in \mathbb{R} : x^3 + x^2 - x - 1 = 0\}$

①

- - - (取交)

intersection

① Def.: $A \cap B = \{x : x \in A \wedge x \in B\}$

$(x \in A \cap B \Leftrightarrow x \in A \wedge x \in B)$

disjoint

Def.: Sets A and B are <u>disjoint</u>

$\Leftrightarrow A \cap B = \emptyset$

union

② Def.: $A \cup B = \{x : x \in A \vee x \in B\}$

$(x \in A \cup B \Leftrightarrow \cdots)$

、 ~ ~ (取交)

2

$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$

$A^c = U \setminus A$

$(A)^c = A$



Thm.: $\forall$ sets A, B, C

① $\phi \cup A = A$,

$\phi \cap A = \phi$

② $A \cap B \subseteq \underbrace{A}_{(\text{or } B)} \subseteq A \cup B$

Proof:

1. $\forall x \in A \cup B,\ x \in A$

$\therefore A \subseteq B \Rightarrow \begin{cases} A \cap B = A \\ A \cup B = B \end{cases}$

Proof: Let [↓]

1. $B \subseteq A \cup B$

2. $\forall x \in A \cup B$

case1: If $x \in A$, $A \subseteq B \Rightarrow x \in B$

case2: If $x \in B$ ... ①

③ $A \cup B = B \cup A$,

$A \cap B = B \cap A$

④ $A \cup (B \cup C) = (A \cup B) \cup C$,

$\dots \cap (\dots \cap \dots) = \dots \dots$

⑤ $A \subseteq B \Rightarrow \begin{cases} (A \cup C) \subseteq (B \cup C) \\ (A \cap C) \subseteq (B \cap C) \end{cases}$

Proof: ... (見hw2)

Thm.:

$\forall$ universe U, $\forall$ sets $A, B \subseteq U$,

$$\begin{cases} (A \cap B)^c = A^c \cup B^c \\ (A \cup B)^c = A^c \cap B^c \end{cases}$$

Proof: ~~~ (★) (R 是 B)

$A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$

Distributive laws

① $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

② $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof: (of ①)

1. (證 $\subseteq$ ...)

let $x \in A \cup (B \cap C)$

case 1: $x \in A$

$\Rightarrow \begin{cases} x \in A \cup B \\ x \in A \cup C \end{cases} \Rightarrow x \in D_2$

case 2: $x \in B \cap C$

$\Rightarrow \begin{cases} x \in B \Rightarrow x \in A \cup B \\ x \in C \Rightarrow x \in A \cup C \end{cases} \Rightarrow x \in D_2$

2. (證 ... $\supseteq$ ...)

let $x \in (A \cup B) \cap (A \cup C)$

(自己---(见证)

$$A = (A \setminus B) \cup (A \cap B)$$

(方法1:) Proof:

(自己---(见证))

(方法2:)

---(见证)

$$A \subseteq B \iff B^c \subseteq A^c$$

Proof: ---(见证)



function

Def.: $\forall$ sets A, B,
domain → codomain

a func. f from A to B ($f: A \to B$)

is a rule which assigns to every element $a \in A$ exactly one element $b \in B$, $f(a) = b$

Image

Let set $U \subseteq A$, func. $f: A \to B$

the image of U is: $f(U) = \{f(x) \in B : x \in U\}$

Range

range of f is: $\text{range}(f) = f(A)$

Inverse image

Let set $V \subseteq B$, func. $f: A \to B$

the inverse image of U is: $f^{-1}(V) = \{a \in A : f(a) \in V\}$

Injective (又: one-to-one)

Def.: func. $f: A \to B$ is inj.

$\Leftrightarrow \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2$

$\Leftrightarrow \forall a_1, a_2 \in A,\ a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$

Surjective (又: onto)

Def.: func. $f: A \to B$ is surj.

$\Leftrightarrow f(A) = B$

$\Leftrightarrow \forall b \in B, \exists a \in A$, s.t. $f(a) = b$

Bijective (又: 一一 ①)

Def.: ——— bij. $\Leftrightarrow$ { f is inj. / f is surj.

Composition

$f: A \to B$, $g: B \to C$

$g \circ f : A \to C$

$\forall a \in A, g \circ f(a) = g(f(a))$

Thm.: ① $\begin{cases} f \text{ is inj.} \\ g \text{ is inj.} \end{cases} \Rightarrow g \circ f$ is inj.

② $\begin{cases} f \text{ is surj.} \\ g \text{ is surj.} \end{cases} \Rightarrow g \circ f$ is surj.

③ $\begin{cases} f \text{ is bij.} \\ g \text{ is bij.} \end{cases} \Rightarrow g \circ f$ is bij.

Proof: (of ①)

① ... (見黑)